



CS 340: Algorithms and Data Structures

Sorting

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Objectives

- Recognize how different sorting algorithms implement the same process with different algorithms
- Recognize the general algorithm for a number of algorithms, i.e., selection sort, merge sort, or quick sort
- Compute the Big-O runtimes of the algorithms

Why Sorting?

- Processing data when knowing the ordering may be critical for the proper functioning of the algorithm.
- Keeping data sorted can reduce search time
- Having data sorted in some order has immense impact on how long it might take to do something.
- It allows us to present information to the user in a way that makes sense to the user.
- It smooths out the user interface. There is no possible way we could use computers without it.

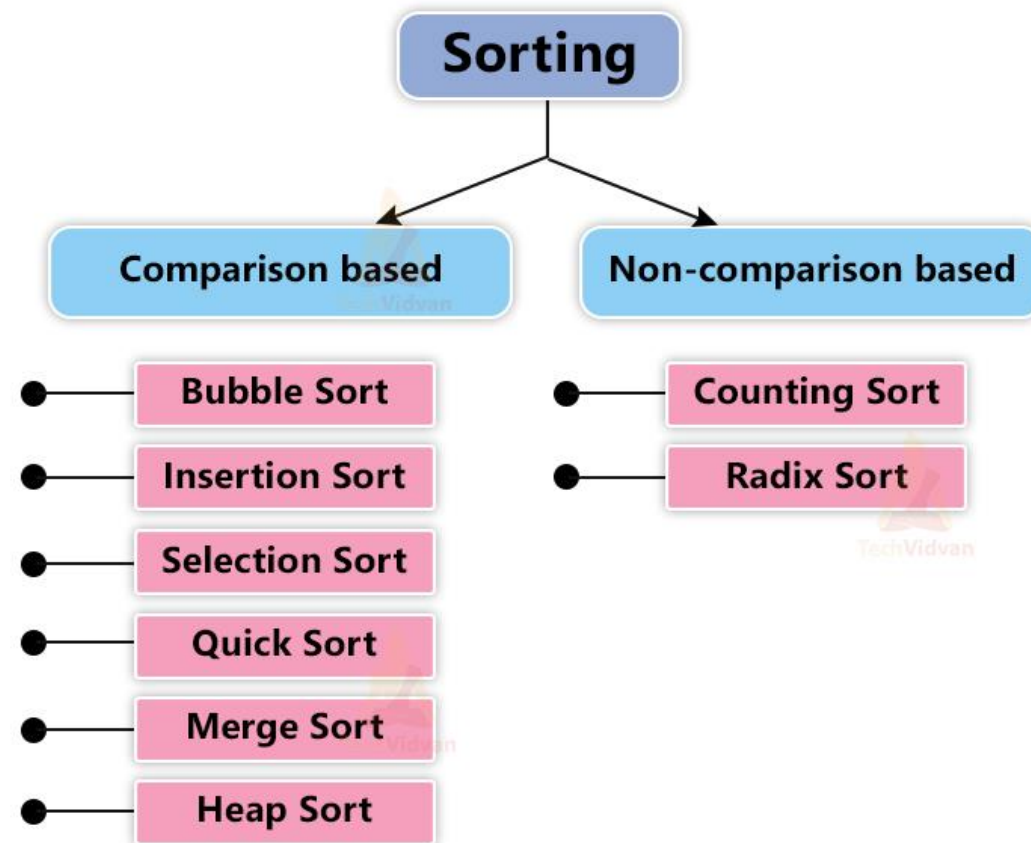
Examples

- •The contact list in your phone is sorted, which means you can easily access your desired contact from your phone since the data is arranged in that manner for you. In other words, “it is sorted”.
- While shopping on flip kart or amazon, you sort items based on your choice, that is, price low to high or high to low.
- Bubble sorting is used in programming TV to sort channels based on audience viewing time!
- Databases use external merge sort to sort sets of data that are too large to be loaded entirely into memory

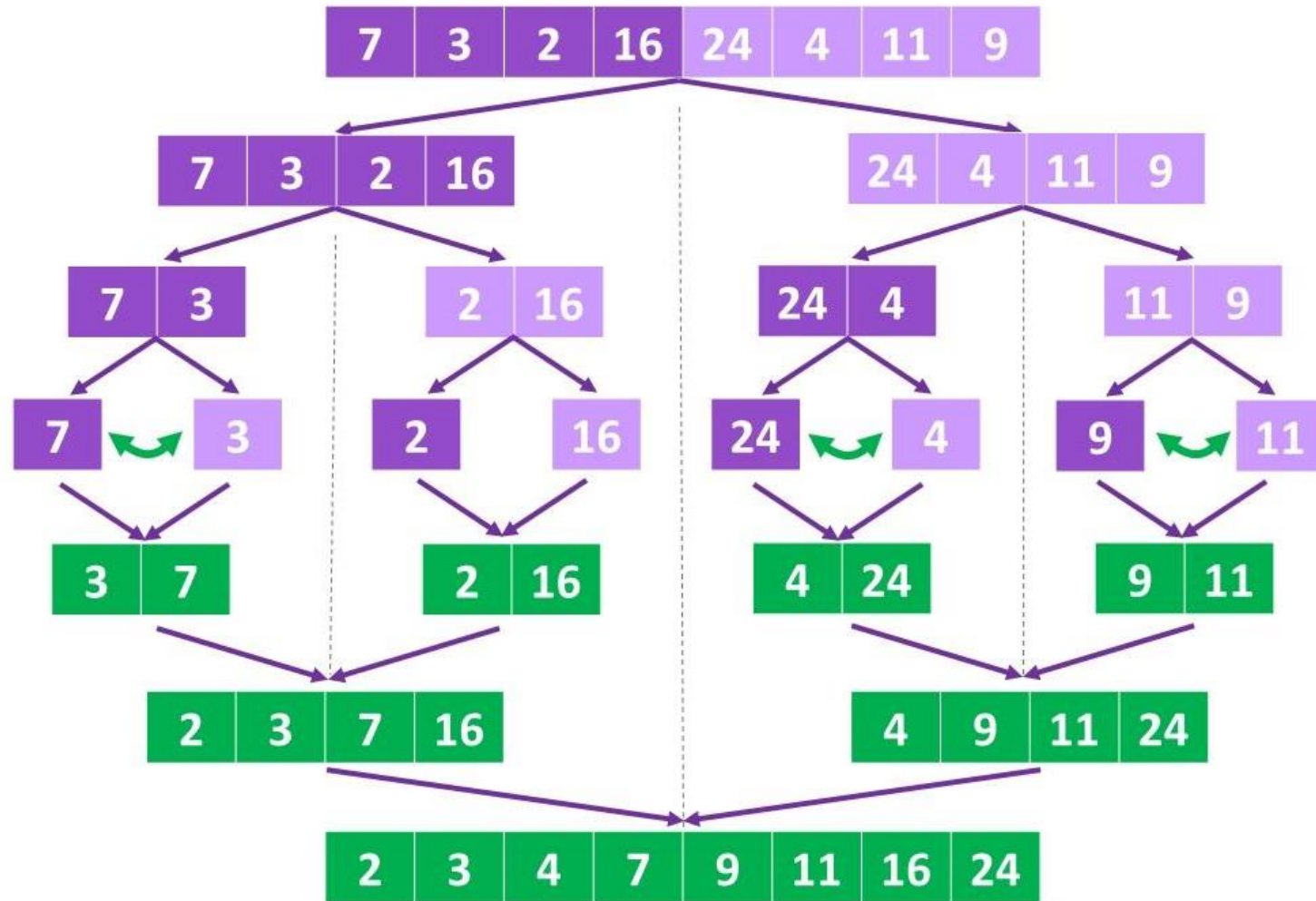
Search Algorithms Benefit from Sorting

- We use search algorithms a lot in computer science. Just think of how many times a day you use Google, or search for a file on your computer.
- We've determined that search algorithms work better when the items they search over are sorted. Can we write an algorithm to sort items efficiently?

For example...



Merge Sort

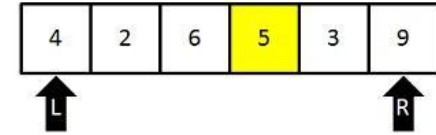


Quick Sort Example

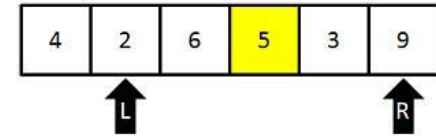
Step 1
Determine pivot



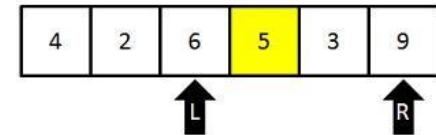
Step 2
Start pointers at left and right



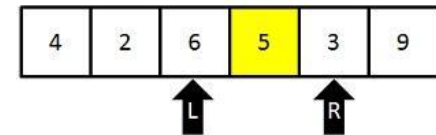
Step 3
Since $4 < 5$, shift left pointer



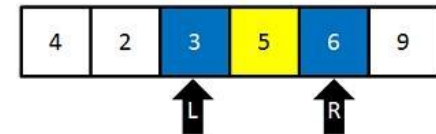
Step 4
Since $2 < 5$, shift left pointer
Since $6 > 5$, stop



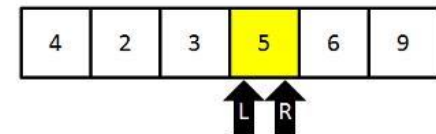
Step 5
Since $9 > 5$, shift right pointer
Since $3 < 5$, stop



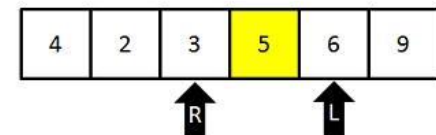
Step 6
Swap values at pointers



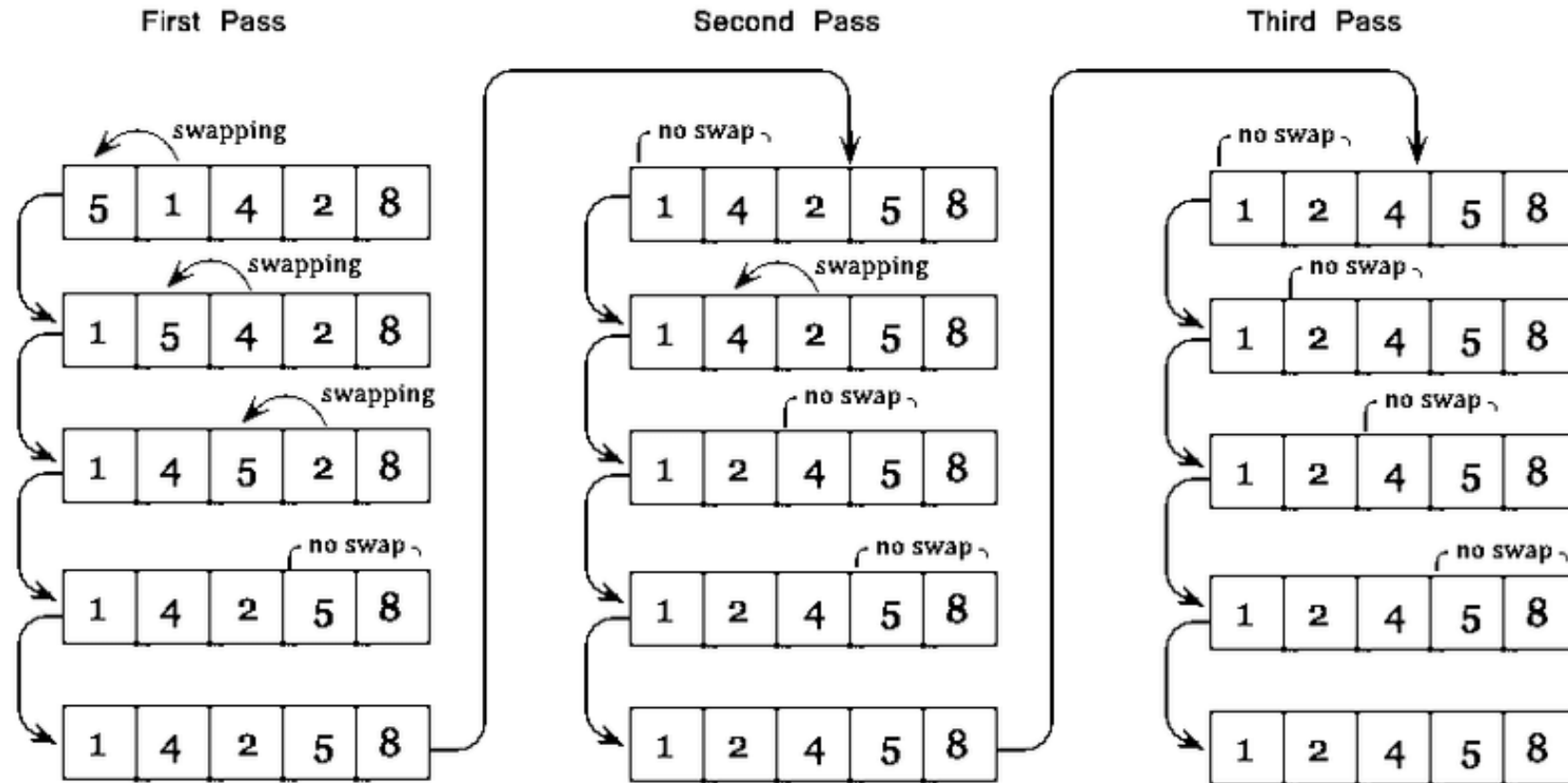
Step 7
Move pointers one more step



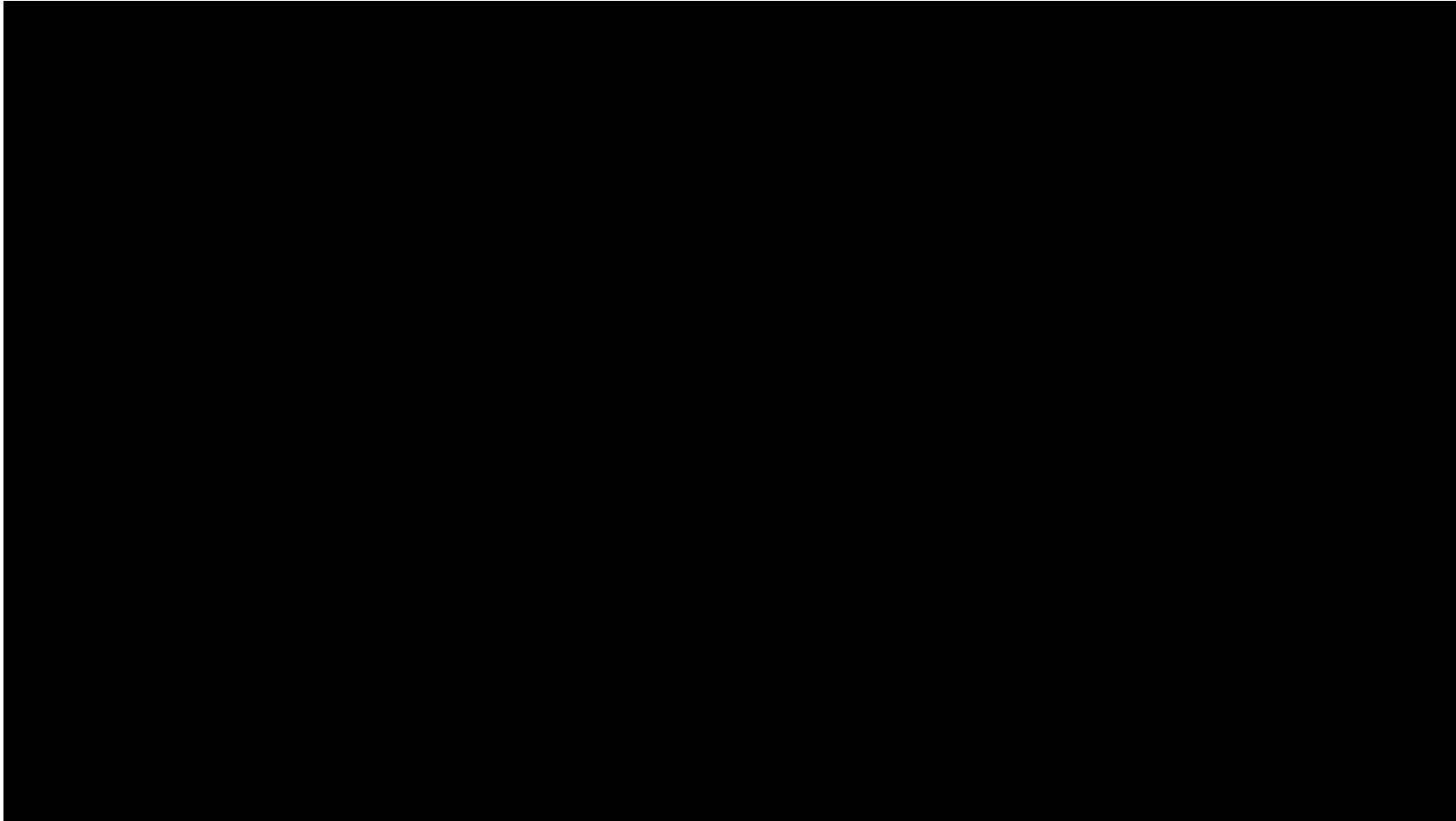
Step 8
Since $5 == 5$,
move pointers one more step
Stop



Bubble Sort



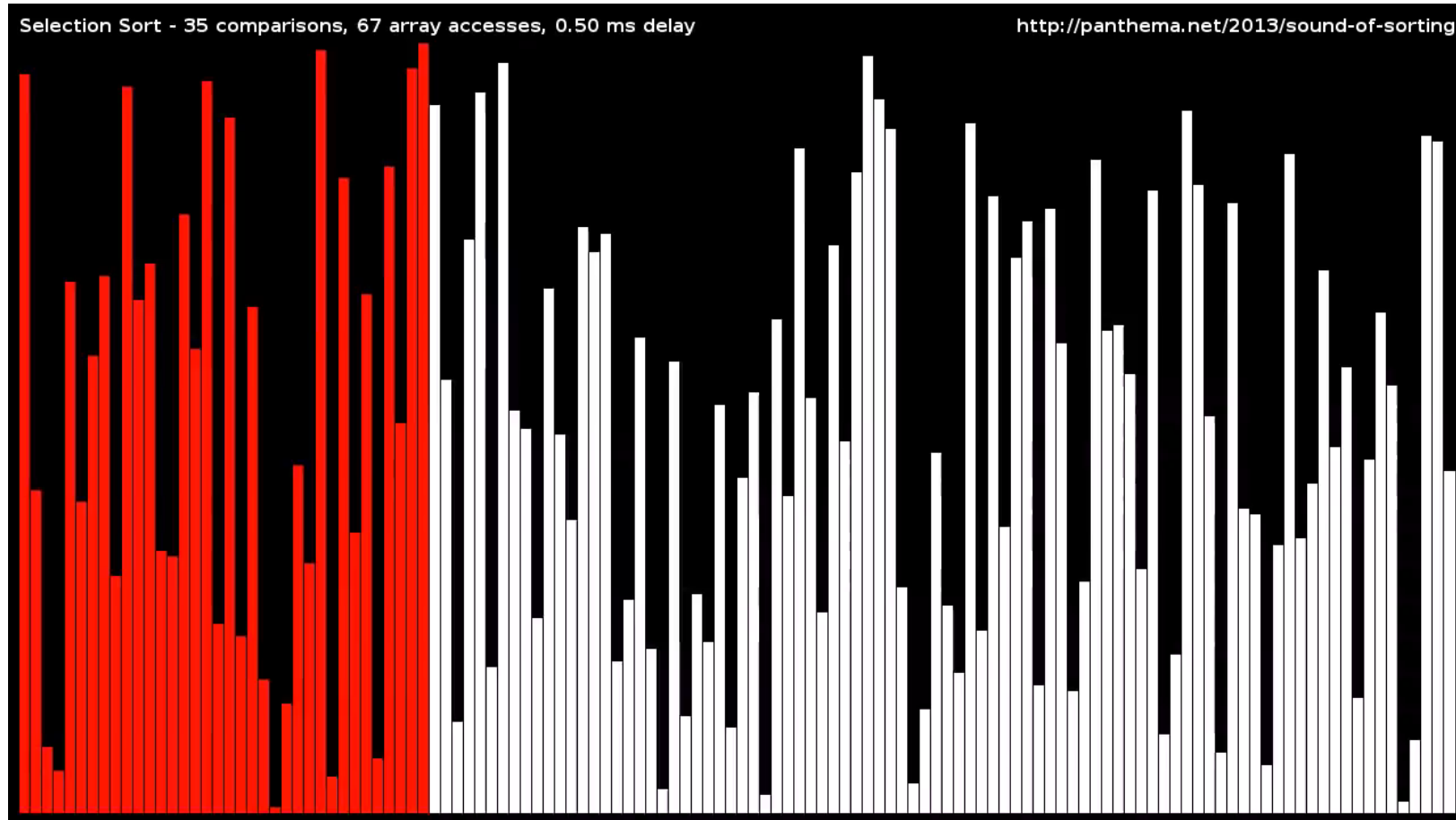
Video



Many Ways of Sorting

- There are a ton of algorithms that we can use to sort a list.
- We'll use <https://visualgo.net/bn/sorting> to visualize some of these algorithms.

Another interesting video



Examples and codes

- <https://users.monash.edu/~lloyd/tildeAlgDS/Sort/>